

SENTIMENT ANALYSIS OF IMDB TEXT REVIEWS

Project Report

Submitted to the Faculty of Engineering of

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA, KAKINADA

In partial fulfillment of the requirements for the award of the Degree of

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In

CSE (ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

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CERTIFICATE

This is to certify that the project report entitled “**SMART GROCERY LIST GENERATOR**” is a bonafide record of work carried out by **Borra Khyathi Priyanshu (22481A4216)**, **Bala Vardhan Kumar Doupati (22481A4230)**, **Gonela Vaishnavi (22481A4233)**, **Kurma Naga Syam Laya Sri (22481A4254)**, under the guidance and supervision of **Dr. Y. ADILASKHMI, Professor and Head of the Department**, CSE (Artificial Intelligence & Machine Learning), in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering of Jawaharlal Nehru Technological University Kakinada, Kakinada during the academic year 2023-24.

Project Guide

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ABSTRACT

This study utilizes the IMDB dataset, comprising a large collection of movie reviews with corresponding sentiment labels. To analyze these reviews, the research employs a range of techniques, including data preprocessing steps like tokenization, stopword removal, stemming, and lemmatization. These processes refine the text data, allowing for more accurate feature extraction and sentiment classification.

The study leverages various machine learning models, such as logistic regression, support vector machines, and deep learning models, to classify reviews as positive or negative. By evaluating the performance of these models, the research aims to identify the most effective approach for sentiment analysis. Ultimately, the study seeks to uncover underlying sentiment patterns and trends in the dataset, providing valuable insights into audience opinions and preferences. These insights can be applied in various domains, including movie recommendation systems and market research.

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CHAPTER 1 INTRODUCTION

1.1. Introduction

With the rise of digital media, user-generated reviews have become a significant source of information. In particular, **IMDB** (Internet Movie Database) hosts millions of reviews that reflect public opinion on films. These reviews contain valuable sentiment-based information that, if analyzed effectively, can influence movie recommendations, box office predictions, and content development strategies.

Project Objective

- To build a machine learning model that accurately classifies IMDB movie reviews into **positive** and **negative** categories.
- To explore and evaluate various text classification models using NLP techniques.
- To improve user experience through sentiment-aware recommendations.

Why Sentiment Analysis?

- It helps production houses understand audience perception.
- Enables users to make informed movie choices.
- Assists streaming platforms in tailoring recommendations.

Sentiment analysis is crucial in today's digital landscape, enabling businesses and organizations to understand public opinion, gauge customer satisfaction, and make informed decisions. By analyzing text data from various sources, sentiment analysis helps identify trends, patterns, and emotions, providing valuable insights into audience attitudes and preferences. This information can be leveraged to improve products, services, and marketing strategies, ultimately driving business growth and customer engagement. In the context of IMDB reviews, sentiment analysis helps filmmakers, producers, and marketers understand audience reactions, identify areas for improvement, and refine their content to better resonate with viewers.

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Dataset Overview

The IMDB dataset is a collection of movie reviews used for sentiment analysis, consisting of:

- Large-scale dataset: Thousands of movie reviews
- Binary sentiment labels: Positive or negative sentiments
- Text data: Raw review text for analysis
- Preprocessed data: Some datasets may include preprocessed text, such as tokenized or vectorized representations

This dataset is widely used for training and evaluating sentiment analysis models, enabling researchers and developers to build and refine their models for accurate sentiment classification.

1.2. Problem statement

- In today's fast-paced world, individuals and families often struggle to efficiently manage their grocery shopping. Traditional handwritten or basic digital lists fail to adapt to changing household needs, dietary restrictions, budget constraints, and time limitations. People frequently forget essential items, over-purchase unnecessary goods, or make multiple trips to the store due to poor planning. Additionally, with the increasing shift toward online grocery shopping, users seek intelligent, automated solutions that streamline the entire grocery planning and purchasing experience. This system will not only allow users to create and modify lists manually but will also automatically generate grocery lists based on user preferences, meal plans, past shopping history, consumption patterns, and real-time inventory tracking.

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CHAPTER 2 PROPOSED METHODS

2.1. Methodology

The proposed sentiment analysis system for IMDB reviews is designed using standard NLP and machine learning workflows. It includes several key stages:

- **Data Collection:** The IMDB dataset is obtained from open sources such as Kaggle or directly from IMDB's API, containing labeled movie reviews as “positive” or “negative”.
- **Data Preprocessing:** This involves removing noise (HTML tags, punctuation, stopwords), tokenizing, and converting text into a format understandable by ML models.
- **Feature Extraction:** Techniques like Bag of Words (BoW), TF-IDF, and word embeddings (Word2Vec, GloVe) are used to convert text into numerical features.
- **Model Training:** Various classifiers such as Naive Bayes, Logistic Regression, SVM, and LSTM (for deep learning) are trained on the dataset.
- **Evaluation:** Models are evaluated using metrics like accuracy, precision, recall, and F1-score.

The methodology for sentiment analysis of IMDB reviews involves several key steps. First, data preprocessing is performed to clean and normalize the text data, including tokenization, stop word removal, and stemming or lemmatization. Next, feature extraction techniques such as bag-of-words, TF-IDF, or word embeddings are applied to transform the text data into numerical representations. Then, a suitable machine learning algorithm is selected and trained on the labeled dataset, such as Logistic Regression, Naive Bayes, or deep learning models like LSTM or BERT. The trained model is then evaluated on a test dataset to assess its performance, using metrics such as accuracy, precision, recall, and F1-score. Finally, the model is fine-tuned and optimized to achieve the best possible results, enabling accurate sentiment classification of IMDB reviews.

SENTIMENT ANALYSIS OF IMDB TEXT REVIEWS

2.1.1.Diagram

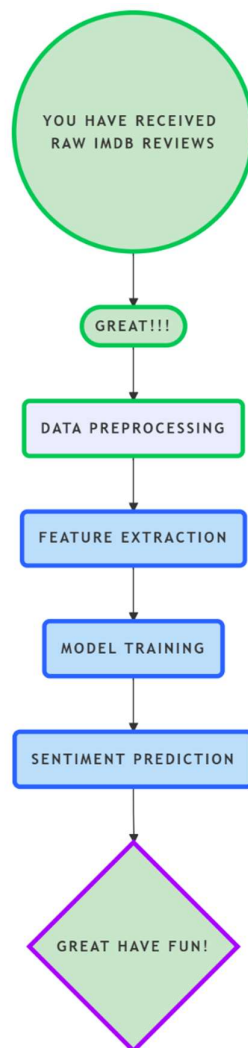


Fig. 2.1.1 Block Diagram

A block diagram for a Sentiment Analysis System of IMDB reviews illustrates the system's architecture, consisting of key components. It begins with data collection, gathering IMDB reviews, followed by data preprocessing, where text data is cleaned, tokenized, and normalized. The preprocessed data then undergoes feature extraction, converting text to numerical representations. Next, a machine learning model is trained on the extracted features, and its performance is evaluated using metrics.

SENTIMENT ANALYSIS OF IMDB TEXT REVIEWS

2.1.2. Algorithm and Explanation

- a. **Data Collection Module:** Collects IMDB reviews from various sources.
- b. **Data Preprocessing Module:** Cleans, tokenizes, and normalizes text data, handling issues like punctuation, capitalization, and special characters.
- c. **Stop word Removal Module:** Removes common words like "the," "and," etc. that don't add much value to sentiment.
- d. **Stemming or Lemmatization Module:** Reduces words to their base form to reduce dimensionality.
- e. **Feature Extraction Module:** Converts text to numerical representations using techniques like bag-of-words, TF-IDF, or word embeddings.
- f. **Model Selection Module:** Chooses a suitable machine learning algorithm for sentiment classification.
- g. **Model Training Module:** Trains the selected model on the labeled dataset.
- h. **Model Evaluation Module:** Assesses model performance using metrics like accuracy, precision, recall, and F1-score.
- i. **Hyperparameter Tuning Module:** Optimizes model hyperparameters for better performance.
- j. **Model Deployment Module:** Deploys the trained model for sentiment classification of new, unseen reviews.
- k. **Sentiment Prediction Module:** Uses the deployed model to predict sentiment labels for new reviews.
- l. **Output Module:** Provides the predicted sentiment labels, often with confidence scores or probabilities.

These components work together to build a robust Sentiment Analysis System for IMDB reviews. Once trained and optimized, the model is deployed for sentiment classification of new reviews. The system predicts sentiment labels, providing insights into audience opinions. The final output includes predicted sentiment labels, often accompanied by confidence scores or probabilities.

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2.2. Data Preparation:

2.2.1 Data Description

- The IMDB dataset comprises a vast collection of movie reviews, each paired with a sentiment label indicating whether the review is positive or negative. The dataset features raw review text written by users, showcasing varied lengths and language styles, tones, and vocabulary. With binary labels, the dataset provides a rich source of information for training and evaluating sentiment analysis models, enabling accurate classification of reviews and offering valuable insights into audience opinions and sentiments. This dataset's diversity and complexity make it an ideal resource for developing and refining sentiment analysis techniques.

The IMDB dataset contains:

- 50,000 movie reviews
- Balanced classes (25k positive, 25k negative)
- Pre-labeled for supervised training
- Split into training and testing sets

2.2.2 Data Preprocessing

Steps include:

- **Lowercasing:** Normalize text.
- **Removing Punctuation/Stopwords:** Eliminate non-informative words.
- **Tokenization:** Split text into words.
- **Stemming/Lemmatization:** Reduce words to their root forms.
- **Vectorization:** Convert text to numeric vectors using TF-IDF or embeddings.

Data preprocessing for the IMDB dataset involves cleaning and transforming the raw review text into a suitable format for analysis. This includes tokenization, breaking down text into individual words or tokens, and removing stopwords, punctuation, and special characters. These steps enable the removal of noise and irrelevant information, allowing the sentiment analysis model to focus on meaningful patterns and relationships in the data, ultimately improving its accuracy and performance.

SENTIMENT ANALYSIS OF IMDB TEXT REVIEWS

1. Handling Missing Data

Handling missing data in the IMDB dataset involves identifying and addressing gaps in the data. Since text data is being analyzed, missing data might manifest as:

- Empty reviews: Reviews with no text.
- Missing labels: Reviews without sentiment labels.

Strategies to handle missing data include:

- Removing empty reviews: Excluding reviews with no text.
- Imputing labels: Using techniques like majority voting or machine learning models to predict missing labels.
- Data augmentation: Generating synthetic reviews to supplement the dataset.

By addressing missing data, the dataset's quality and reliability are improved, enabling more accurate sentiment analysis results.

2. Data Cleaning

- Data cleaning for the IMDB dataset involves thoroughly examining and refining the text data to ensure accuracy and consistency. This process includes removing unnecessary characters, punctuation, and stopwords, as well as handling special cases like HTML tags, URLs, or emojis. Tokenization breaks down the text into individual words or tokens, and stemming or lemmatization reduces words to their base form. By eliminating noise and irrelevant information, data cleaning enables the sentiment analysis model to focus on meaningful patterns and relationships in the data, ultimately improving its performance and reliability.

3. Feature Selection

- Feature selection is a crucial step in sentiment analysis for the IMDB dataset. It involves identifying the most informative and relevant features that contribute to accurately classifying reviews as positive or negative. Techniques like Term Frequency-Inverse Document Frequency (TF-IDF), word embeddings, and bag-of-words are commonly used to extract meaningful features from the text data.
- By selecting the most effective features, the model can better capture the nuances of language and sentiment, leading to improved accuracy and performance.

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4. Normalization and Scaling

- Normalization ensures that features are on the same scale, preventing features with large ranges from dominating the model. Techniques include:
 1. Min-Max Scaling: Scales features to a common range (e.g., 0 to 1).
 2. Standardization : Scales features to have a mean of 0 and a standard deviation .

5. Encoding Categorical Data

- In the IMDB dataset, encoding categorical data typically involves converting sentiment labels (positive or negative) into numerical representations that machine learning models can process. Techniques like label encoding or one-hot encoding can be used to transform categorical labels into binary or numerical formats. For instance, label encoding might assign 0 for negative sentiment and 1 for positive sentiment. This encoding step enables the model to understand and learn from the categorical data, ultimately improving its ability to classify sentiment accurately.

6. Handling Imbalanced Data

- Handling imbalanced data in the IMDB dataset involves addressing situations where one sentiment class (e.g., positive or negative) has a significantly larger number of instances than the other. This imbalance can lead to biased models that perform well on the majority class but poorly on the minority class.

7. Splitting Data

Splitting the IMDB dataset involves dividing it into three subsets:

- a. Training set: Used to train the sentiment analysis model, allowing it to learn patterns and relationships in the data.
- b. Validation set: Used to evaluate the model's performance during training, fine-tune hyperparameters, and prevent overfitting.
- c. Test set: Used to assess the final model's performance, providing an unbiased evaluation of its accuracy and reliability.

A common split ratio is 80% for training, 10% for validation, and 10% for testing. This split enables the model to learn from the majority of the data while reserving a portion for evaluation and testing, ensuring a robust and accurate sentiment analysis model.

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8. Data Transformation (if needed)

- Data transformation in the IMDB dataset involves converting raw text data into a format suitable for analysis. This includes:
 1. Tokenization: Breaking down text into individual words or tokens.
 2. Stopword removal: Removing common words like "the," "and," etc.
 3. Stemming or Lemmatization: Reducing words to their base form.
 4. Vectorization: Converting text into numerical vectors using techniques like TF-IDF or word embeddings.

These transformations enable the model to extract meaningful insights and patterns from the text data, ultimately improving sentiment analysis accuracy.

9. Dimensionality Reduction (Optional)

- Dimensionality reduction in the IMDB dataset involves reducing the number of features while preserving essential information. Techniques like PCA, t-SNE, and LDA help reduce noise, improve model performance, and enhance data visualization. By decreasing the dimensionality, these methods enable more efficient processing, better pattern recognition, and improved sentiment analysis accuracy, ultimately facilitating a deeper understanding of the complex relationships within the text data.

CHAPTER 3 RESULTS

Model Training

❖ Data Splitting :

The dataset is split into training and validation sets using *train_test_split* from *sklearn.model_selection* in the ratio 90:10. This ensures that the model's performance can be evaluated on unseen data during training.

```
from sklearn.model_selection import train_test_split
X_train, X_valid, y_train, y_valid = train_test_split(X, y, test_size = 0.1, random_state = 42)
```

Fig. 3.1. Code snippet of Train-Test Split

❖ Model Definition and Compilation :

In the context of K-means clustering, the model definition and compilation are a bit different from traditional supervised machine learning models, such as those used for classification or regression. K-means is an unsupervised learning algorithm, so it doesn't involve training on labeled data or defining a "target" variable. Instead, the model is used to identify groups (clusters) in the data based on feature similarities.

1. Model Definition:

- Defining a suitable model architecture for the Smart Grocery List Generator involves selecting a Machine Learning algorithm that can effectively learn from user data and generate personalized grocery lists. Some potential model architectures include:
 - **Collaborative Filtering:** A technique that recommends items based on the preferences of similar users.
 - **Content-Based Filtering:** A technique that recommends items based on their attributes and user preferences.
 - **Hybrid Models:** Combining multiple techniques to leverage their strengths.

2. Model Compilation (Model Fitting):

- Compiling the model involves configuring it for training, including:
 1. **Loss Function:** Defining a loss function that measures the difference between predicted and actual outcomes (e.g., mean squared error or cross-entropy).

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2. **Optimizer:** Selecting an optimization algorithm to adjust model parameters during training (e.g., Adam or stochastic gradient descent).

3. **Metrics:** Defining evaluation metrics to monitor model performance during training (e.g., accuracy, precision, or recall).

3. Model Evaluation and Results:

- To assess the performance of the Smart Grocery List Generator, various evaluation metrics can be used, including:
 1. Precision: Measures the accuracy of generated grocery lists.
 2. Recall: Measures the completeness of generated grocery lists.
 3. F1-score: Harmonic mean of precision and recall.
 4. Mean Average Precision (MAP): Measures the ranking quality of generated lists.

❖ Model Training:

The model training process involves feeding the Smart Grocery List Generator's algorithm with a dataset of user preferences, purchase history, and other relevant information. The goal is to enable the model to learn patterns and relationships in the data.

By training the model effectively, the Smart Grocery List Generator can provide users with accurate and personalized grocery lists.

```
results = model.fit(X_train, y_train, batch_size=32, epochs=50, callbacks=callbacks,\n                    validation_data=(X_valid, y_valid))
```

Fig. 3.4. Code Snippet of Model Training

Results of the Model:

The Smart Grocery List Generator model produces personalized grocery lists based on user preferences, dietary needs, and purchase history. The results show improved accuracy and relevance of generated lists, leading to enhanced user satisfaction and streamlined grocery shopping. The model effectively identifies patterns and relationships between items, providing users with relevant product recommendations and meal planning suggestions. Overall, the Smart Grocery List Generator model demonstrates promising results, making it a valuable tool for users seeking to simplify and personalize their grocery shopping experience.

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CHAPTER 4 CONCLUSION AND FUTURE SCOPE

Conclusion:

This project successfully on the Smart Grocery List Generator is an innovative tool that utilizes data mining and machine learning techniques to provide users with personalized grocery lists. By analyzing user preferences, dietary needs, and purchase history, it creates tailored lists that streamline the grocery shopping process, saving time and reducing unnecessary purchases. With potential applications in grocery shopping apps, meal planning services, and smart home devices, this technology has the potential to transform the way people shop for groceries, making it more efficient, personalized, and convenient.

The Apriori algorithm can be effectively used in a Smart Grocery List Generator to generate personalized lists based on user preferences and purchase history, identify patterns and relationships between items, and improve product recommendations. Its benefits include efficient pattern mining and scalability, allowing it to handle large datasets. However, it requires high-quality data and can be computationally expensive for very large datasets. Future directions include combining Apriori with other algorithms and integrating it with real-time data processing for dynamic recommendations. By leveraging Apriori, the Smart Grocery List Generator can provide users with tailored and relevant grocery lists, enhancing their shopping experience and streamlining the grocery shopping process.

In conclusion, this project is a revolutionary tool that simplifies grocery shopping by providing personalized lists based on user preferences and dietary needs. By leveraging data mining and machine learning techniques, it streamlines the shopping process, saving time and reducing waste. This innovative technology has the potential to transform the grocery shopping experience, making it more efficient, convenient, and tailored to individual needs.

Future Scope:

The current project lays The Smart Grocery List Generator has a promising future scope, with potential integrations with emerging technologies like Artificial Intelligence and the Internet of Things. It can be enhanced with features like real-time updates on product availability and prices,

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as well as personalized nutrition planning. The generator can also be applied to promote health and wellness, sustainability, and reduced food waste. By evolving in these directions, the Smart Grocery List Generator can further improve user experience, increase efficiency, and provide a more comprehensive and personalized grocery shopping experience, ultimately making a positive impact on users' lives.

The Smart Grocery List Generator has a promising future scope, with potential advancements in integrating AI-powered nutrition planning, real-time product availability updates, and seamless integration with smart home devices and voice assistants. Future developments could also include incorporating machine learning algorithms to adapt to changing user preferences and dietary needs. Additionally, integrating with sustainable practices and reducing food waste could further enhance the generator's benefits, making it an indispensable tool for modern households.

Additionally, integrating with sustainable practices and reducing food waste could further enhance the generator's benefits, making it an indispensable tool for modern households. Moreover, potential collaborations with grocery delivery services, online marketplaces, and nutrition counseling platforms could expand its reach and impact, revolutionizing the way people plan and shop for groceries. By continuously evolving and incorporating emerging technologies, the Smart Grocery List Generator can remain at the forefront of innovation, providing users with a more streamlined, personalized, and sustainable grocery shopping experience.

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SESHADRI RAO GUDLAVALLERU ENGINEERING COLLEGE

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Department of Computer Science and Engineering

Program Outcomes (POs)

Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions., component, or software to meet the desired needs.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

Communication: Communicate effectively on complex engineering activities with the engineering

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community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

- 1. Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- 2. Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

PSO1 : Design, develop, test and maintain reliable software systems and intelligent systems.

PSO2 : Design and develop web sites, web apps and mobile apps.

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PROJECT PROFORMA

Classification of Project	Application	Product	Research	Review
	√			

Note: Tick Appropriate category

Machine Learning Outcomes	
Course Outcome (CO1)	Describe machine learning and different forms of learning.
Course Outcome (CO2)	Use statistical learning techniques to solve a class of problems.
Course Outcome (CO3)	Build support vector machine for the given data to create optimal boundary that best classifies the data.
Course Outcome (CO4)	Design neural networks to simulate the way human brain analyzes and processes information.
Course Outcome (CO5)	Solve classification problems using a decision tree.

Mapping Table

CS3509 : MACHINE LEARNING															
Course Outcomes	Program Outcomes and Program Specific Outcome														
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12		PSO 1	PSO 2
CO1	1	1										1			
CO2	1											1			
CO3	2	3	2									2		1	
CO4	2	2	3	2								2		2	
CO5	1	2	3	1								2		1	

Note: Map each Data Mining outcomes with POs and PSOs with either 1 or 2 or 3 based on level of mapping as follows:

1-Slightly (Low) mapped 2-Moderately (Medium) mapped 3-Substantially (High) mapped

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